

Exact Estimation of Exponential Family Models on Massive Mixed Categorical–Continuous Data via Tilted Cell Moments

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July 2026

Abstract

When every covariate of a generalized linear model is categorical, the log likelihood collapses onto within-cell aggregates of the cross-classification, and estimation on the J cells reproduces the microdata maximum-likelihood estimator at a fraction of the cost. A single continuous covariate breaks it: we show that a fixed-dimensional cell summary can no longer reproduce the likelihood *function*, because doing so would require the entire within-cell empirical distribution of the continuous covariates. The contribution of this paper is a distinction between likelihood exactness, which is impossible, and *score* exactness, which is not: a Newton or Fisher-scoring step needs only the gradient and curvature at the current iterate, and for exponential dispersion families these are, exactly, linear combinations of three low-order moments of an *exponentially tilted* within-cell measure—for the Poisson family, an Esscher transform of the within-cell exposure measure—that one pass computes. Consequently every Newton or Fisher-scoring iteration is computable from one streaming pass over the microdata at $O(Np_c^2)$ cost, with p_c the number of continuous covariates only, followed by algebra on the cells; the resulting iterates coincide with full-design Newton up to floating-point reordering, so the method is exact rather than an approximation. We give the streaming algorithm, its complexity, memory, and numerical-stability properties, and exact cell-accumulated robust and cluster-robust covariance estimators. Experiments confirm agreement with dense Newton to 10^{-13} – 10^{-15} across Poisson, logistic, and gamma models, and benchmark against the implementations practitioners use: one to two orders of magnitude faster and 12–50× lighter in memory than `statsmodels`, faster than `glum` and a sparse-design IRLS, and—for absorbed high-dimensional fixed effects—about twice as fast and several times lighter than `pyfixest` while returning the exact estimate, with $O(N)$ time and memory verified to 5×10^7 records and accuracy preserved across a battery

*Swedish Institute for Social Research (SOFI), Stockholm University, SE-106 91 Stockholm, Sweden. E-mail: andreas.ljungstrom@sofi.su.se. Reproduction code for all numerical results is available at <https://github.com/andreasljungstrm/stpois> (directory `paper/stats/code`). An online supplement contains the companion theory (convergence of nonlinear fixed-effect absorption and its rates, high-dimensional asymptotics, Firth and penalized extensions) and its proofs and simulations.

of ill-conditioned designs. Two applications on public and register-scale data—a logistic on-time model for the 336,776 flights of `nycflights13`, agreeing with `pyfixest` to 10^{-9} once 3,692 aircraft are absorbed, and a five-million-episode epidemiology study with 10,000 absorbed municipality–year effects—fit with inference in seconds. An online supplement develops companion theory: global convergence and sharp local rates for the nonlinear alternating-projections absorption of high-dimensional fixed effects, asymptotics up to the proportional regime $J/N \rightarrow c$, and Firth, penalized, and conditional-likelihood extensions.

Keywords: generalized linear models; exponential tilting; sufficient statistics; streaming algorithms; high-dimensional fixed effects; register data.

1 Introduction

A large share of applied statistical work now takes place on data sets whose size is set not by a sampling budget but by an administrative process: full-population registers, insurance claim files, electronic health records, transportation and digital trace data. The models fitted to such data are rarely exotic—Poisson regression for event rates, logistic regression for binary outcomes, gamma regression for costs—but their covariate structure has a characteristic shape. A small number of continuous covariates of scientific interest (income, exposure, a polygenic score) appear alongside a rich categorical apparatus: demographic strata such as age band $\times$ sex $\times$ education, operational strata such as carrier $\times$ route $\times$ month, and often fixed-effect factors with thousands of levels. The cross-classification of the categorical covariates partitions the N observations into J cells, and it is the interplay between N , J , and the continuous covariates that determines whether maximum-likelihood estimation is an interactive computation or an overnight batch job.

Two classical observations frame the problem. First, when *every* covariate is categorical, the exponential-family log likelihood depends on the data only through within-cell aggregates; estimation on the J collapsed cells reproduces the microdata maximum-likelihood estimator (MLE) exactly, and the per-iteration cost drops from $O(Nq^2)$ to $O(Jq^2)$ for q parameters. This is the standard device behind register-based rate modeling in epidemiology and demography (Holford, 1980; Laird and Olivier, 1981; Clayton and Hills, 1993). Second, the collapse is fragile: a single continuous covariate destroys it, because the cumulant function of the family is strictly convex, so cell sums of nonlinear transforms of the linear predictor cannot be recovered from cell means. Analysts then face an unattractive choice between discretizing the continuous covariates—introducing bias that is hard to quantify—and abandoning the cell structure entirely, paying the full $O(Nq^2)$ price per iteration even though most of the q parameters are attached to covariates that are constant within cells. General-purpose GLM software pays this price by default; software for high-dimensional fixed effects (Guimarães and Portugal, 2010; Gaure, 2013; Correia et al., 2020; Bergé, 2018; Stammann, 2018) removes *dummy-coded factors* from the design by demeaning, but continuous-by-cell interplay of the kind above is not a demeaning problem, and the cell structure of the categorical part goes unexploited.

This paper develops a resolution that is exact rather than approximate. The starting point is a precise statement of what breaks (Proposition 1): with continuous covariates, any statistic from which the likelihood *function* can be evaluated must determine the entire within-cell empirical distribution of the continuous covariates—the within-cell Laplace transform appears in the likelihood, and Laplace transforms determine measures. No fixed cell summary can therefore support the full likelihood surface. The contribution is that estimation does not need the full surface; it needs score and curvature *at the current iterate*, and these have an exact finite representation. Theorem 1 shows that for exponential dispersion families with canonical link,

the score vector and Fisher information at any parameter value are linear combinations of the zeroth-, first-, and second-order moments of an *exponentially tilted probability measure* on each cell—for the Poisson family, an Esscher transform of the within-cell empirical exposure measure, connecting the construction to classical tilting ideas in saddlepoint approximation and empirical likelihood (Esscher, 1932; Daniels, 1954; Efron, 1981; Jensen, 1995; Kitamura and Stutzer, 1997). The conceptual pivot is a distinction between *likelihood exactness* and *score exactness*. Evaluating the likelihood function everywhere would require the full within-cell distribution (Proposition 1); but a Newton or Fisher-scoring step never needs the function everywhere—it needs the gradient and curvature at one point, the current iterate. Those are *score exact* from a fixed-dimensional summary even though the likelihood is not, and the summary that delivers them is the tilt, which moves with the iterate. A fixed collapse fails because it tries to be likelihood exact; the tilted collapse succeeds because it only needs to be score exact, iterate by iterate. Corollary 1 converts the identity into complexity: each Newton iteration costs one streaming $O(N(p^2 + p))$ pass over the microdata— p the number of continuous columns only—plus $O(J(k^2 + kp))$ cell algebra, against $O(N(p + k)^2)$ for the full design, a saving that grows quadratically in the richness k of the categorical structure. Because the identity is exact rather than a linearization, the accelerated iterates coincide with full-design Newton up to floating-point reordering, which Section 5.1 verifies at the 10^{-13} – 10^{-15} level, iterate by iterate, across Poisson, logistic, and gamma models. Fisher scoring under non-canonical links has the same structure (Proposition 2); for the gamma–log family the tilt weights are identically one and the moments are constant across iterations.

The second half of the paper is deliberately computational. Section 4 gives the streaming algorithm and its implementation envelope: numerical safeguards (step-halving, log-domain evaluation, rank-deficiency fallbacks), memory layout and parallelization of the bandwidth-bound accumulation pass, exact cell-accumulated robust and cluster-robust covariance estimators, and the combination with absorbed fixed-effect factors, for which the companion theory—global convergence of the nonlinear alternating-projections cycle, its local rates for two and for three or more factors, and high-dimensional asymptotics up to one fixed effect per two observations—is developed in the online supplement. Section 5 contains the experiments: iterate-level exactness against dense Newton (Section 5.1); time-and-memory benchmarks against the implementations practitioners use—`statsmodels`, a sparse-design IRLS, and `glum` (QuantCo’s optimized categorical-aware GLM) for the non-absorbed model (Section 5.2), and `pyfixest` (the `fixest` algorithm) for the absorbed-factor path (Section 5.3); $O(N)$ time and memory scaling to 5×10^7 records (Section 5.4); and a numerical stability battery (Section 5.6). Section 6 applies the method twice: to the public `nycflights13` data—a logistic on-time-performance model verified against `statsmodels` to 10^{-13} and, with 3,692 absorbed aircraft, against `pyfixest` to 10^{-9} —and to a fully reproducible register-scale synthetic study (five million person-year episodes, 10,000 absorbed municipality–year effects, cluster-robust inference, seconds on one

core). Section 7 summarizes and points to the supplement’s theory. Proofs of the main-paper results are in the appendix; every number regenerates from short scripts included with the manuscript.

Relation to existing computational methods. The idea sits at the intersection of several literatures, and its novelty is best stated by contrast with each. *Collapsed sufficient statistics* for log-linear rate models (Holford, 1980; Clayton and Hills, 1993) are the special case $p = 0$; the tilted moments extend the exact collapse from “all categorical” to “categorical plus continuous,” which is precisely the case the classical collapse cannot handle (Proposition 1). *Matrix-free and Hessian-free Newton methods* (Pearlmutter, 1994; Martens, 2010; Nocedal and Wright, 2006) avoid forming the Hessian by supplying only Hessian–vector products to an iterative linear solver; they reduce memory but still stream the full design through every product and solve the q -dimensional system iteratively and approximately. The tilted representation is different in kind: it forms the *exact* $(p + k)$ -dimensional Hessian and score in closed form from cell moments, so the inner solve is a small direct factorization, not a truncated conjugate-gradient loop, and the reduction is in the dimension of the linear algebra, not merely in whether the matrix is stored. *Grouped or blockwise IRLS* and stochastic second-order methods (Byrd et al., 2011) exploit structure in the working weights or subsample the Hessian; they are approximate where the tilted step is exact, and they do not use the key fact here—that within a cell the information contribution is a tilted moment, computable without revisiting the cell. *Automatic differentiation* (Baydin et al., 2018) would compute the same score and Hessian but by propagating through the N -row design, at $O(N(p + k)^2)$; it mechanizes the derivative, whereas the contribution here is the analytic identity that makes the derivative a cell-level object. *Sparse-design* GLM solvers and out-of-core packages (glum, Balzer et al., 2021; biglm, Lumley, 2011) store the dummy block sparsely; as Section 5.2 shows, this is the strongest general competitor, but the tilted pass never forms a design at all and its cost grows only in the continuous dimension. *Distributed and communication-avoiding* optimization (Zhang et al., 2013; Jaggi et al., 2014) targets the orthogonal axis of splitting one fit across machines; the tilted pass is a single-machine, single-pass reorganization that such schemes could adopt per shard, since its accumulators are additive across data partitions. Finally, the *alternating-projections HDFE* stack (Guimarães and Portugal, 2010; Gaure, 2013; Correia, 2016; Correia et al., 2020; Bergé, 2018) removes dummy factors by demeaning but streams the full non-absorbed design through every IRLS step; the two devices are complementary, and Section 4.5 combines them, with the tilted pass handling the cell-structured part that HDFE tools treat as ordinary covariates. In short, the novelty is neither “a sufficient statistic exists” nor “regroup the sums”: it is the exact, iterate-dependent tilted-moment identity (Theorem 1) that turns the score and the full Hessian of a mixed categorical–continuous GLM into a one-pass cell computation, together with its consequences for scale.

The framework is implemented in the Stata command `stpois` for the Poisson event-history case and in an open-source Python reference implementation used for all experiments here.

2 Exponential family models and the failure of ordinary collapse

2.1 Model and notation

Observations $i = 1, \dots, N$ carry a response y_i , a vector $\mathbf{x}_i \in \mathbb{R}^p$ of *individual-level* covariates (those involving at least one continuous variable, including continuous–categorical interaction columns), a vector $\mathbf{w}_{j(i)} \in \mathbb{R}^k$ of *cell-level* covariates, and a known offset o_i . The map $j : \{1, \dots, N\} \rightarrow \{1, \dots, J\}$ assigns each observation to the cell of the cross-classification of the categorical covariates; $\mathbf{w}_j$ is constant within cell j by construction. Conditional on the covariates, responses are independent draws from an exponential dispersion family

$$f(y_i \mid \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}, \quad (1)$$

with $b : \Theta \rightarrow \mathbb{R}$ strictly convex and three times continuously differentiable on the open canonical parameter space $\Theta \subseteq \mathbb{R}$, and dispersion $\phi > 0$ (McCullagh and Nelder, 1989). The mean and variance are $\mu_i = b'(\theta_i)$ and $\text{Var}(y_i) = \phi b''(\theta_i)$. We model the linear predictor

$$\eta_i = \boldsymbol{\gamma}' \mathbf{x}_i + \boldsymbol{\delta}' \mathbf{w}_{j(i)} + o_i, \quad \boldsymbol{\theta} = (\boldsymbol{\gamma}', \boldsymbol{\delta}')' \in \mathbb{R}^q, \quad q = p + k, \quad (2)$$

and, until Section 3.2, take the link to be canonical, $\theta_i = \eta_i$. The Poisson event-history model that motivates much of the applied work is the special case $b(\theta) = e^\theta$, $\phi = 1$, $y_i \in \{0, 1, 2, \dots\}$ an event count, and $o_i = \log t_i$ the log of the time at risk; the piecewise-exponential equivalence of Holford (1980); Laird and Olivier (1981); Whitehead (1980) makes this the canonical likelihood for register-based survival analysis. Up to terms not involving $\boldsymbol{\theta}$, the log likelihood is

$$\ell(\boldsymbol{\theta}) = \frac{1}{\phi} \sum_{i=1}^N \{y_i \eta_i - b(\eta_i)\}. \quad (3)$$

Because (3) is a sum of concave functions of the linear predictor, ℓ is concave on $\mathbb{R}^q$, strictly so on the quotient of $\mathbb{R}^q$ by the null space of the stacked design $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$. We write $\mathbf{U}(\boldsymbol{\theta}) = \nabla \ell$ for the score, $\mathbf{H}(\boldsymbol{\theta}) = -\nabla^2 \ell$ for the negative Hessian, and note that for canonical links $\mathbf{H}$ equals the Fisher information $\mathbf{F}(\boldsymbol{\theta}) = \mathbb{E}[-\nabla^2 \ell]$, so Newton–Raphson and Fisher scoring coincide.

2.2 Cell collapse and its failure

Suppose first that $p = 0$: every covariate is categorical, so $\eta_i = \boldsymbol{\delta}'\mathbf{w}_j + o_i$ varies over i within a cell only through the offset. For the Poisson family with $o_i = \log t_i$, (3) becomes $\ell(\boldsymbol{\delta}) = \sum_j \{D_j \boldsymbol{\delta}'\mathbf{w}_j - T_j e^{\boldsymbol{\delta}'\mathbf{w}_j}\} + \text{const}$, with $D_j = \sum_{i \in j} y_i$ and $T_j = \sum_{i \in j} t_i$: the collapsed cell data (D_j, T_j) carry all the information, and estimation on the J cells is exact. The following proposition records why no analogous finite collapse survives the introduction of a continuous covariate, and in what precise sense. Define the within-cell *exposure measure* $\nu_j = \sum_{i \in j} e^{o_i} \delta_{\mathbf{x}_i}$ on $\mathbb{R}^p$ (for Poisson, $\nu_j = \sum_{i \in j} t_i \delta_{\mathbf{x}_i}$), a finite discrete measure recording the joint configuration of exposures and continuous covariates in cell j .

Proposition 1 (Collapse and its failure). *Consider the Poisson family with offsets. (i) If $\mathbf{x}_i = \bar{\mathbf{x}}_j$ for all $i \in j$ and all j , the log likelihood (3) depends on the data only through $\{(D_j, T_j, \bar{\mathbf{x}}_j, \sum_{i \in j} y_i \mathbf{x}_i)\}_{j \leq J}$, and the MLE on the collapsed cells equals the microdata MLE. (ii) In general, $\ell(\boldsymbol{\gamma}, \boldsymbol{\delta})$ depends on the data through $\{S_j, D_j, \mathcal{L}_{\nu_j}\}_{j \leq J}$, where $S_j = \sum_{i \in j} y_i \mathbf{x}_i$ and $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) = \int e^{\boldsymbol{\gamma}'\mathbf{x}} \nu_j(d\mathbf{x}) = \sum_{i \in j} t_i e^{\boldsymbol{\gamma}'\mathbf{x}_i}$ is the Laplace transform of ν_j ; and $\mathcal{L}_{\nu_j}$ determines ν_j . Consequently any statistic from which the likelihood function can be computed for all $\boldsymbol{\gamma}$ in an open set must determine the entire within-cell exposure measure, whose support size—hence the dimension of any such statistic—grows with the number of distinct covariate values in the cell. (iii) By strict convexity of the exponential (Jensen), $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \geq T_j e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j}$ with equality for some $\boldsymbol{\gamma} \neq \mathbf{0}$ direction only if $\mathbf{x}$ is ν_j -a.s. constant along that direction: replacing microdata by cell means strictly understates the within-cell mass of the cumulant term and is not a reparametrization of the same likelihood.*

Part (ii) is the formal statement of the practical dictum that “continuous covariates destroy collapsibility”: the fixed-dimensional sufficient statistic $\sum_i y_i \mathbf{a}_i$ for the canonical parameter still exists, but *computing the likelihood*—and hence the MLE—requires functionals of the data whose dimension is of order N , not J . The next subsection shows that the required functionals nevertheless have a special structure: for each trial value of $\boldsymbol{\gamma}$ they are three low-order moments of a tilted measure, computable in one linear pass.

3 Tilted cell moments

3.1 Exponentially tilted cell moments

Definition 1 (Tilted within-cell measures). *Fix $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\delta})$ and let $\omega_i = b''(\eta_i) > 0$ be the information weight of observation i . The information-tilted measure on cell j is the probability measure*

$$P_{j,\boldsymbol{\theta}}(\{i\}) = \frac{\omega_i}{\sum_{i' \in j} \omega_{i'}}, \quad i \in j.$$

For the Poisson family, $\omega_i = \mu_i = t_i e^{\delta' \mathbf{w}_j + \gamma' \mathbf{x}_i}$, the factor $e^{\delta' \mathbf{w}_j}$ cancels, and $P_{j,\boldsymbol{\theta}}$ reduces to the Esscher transform (exponential tilt) of the normalized exposure measure ν_j/T_j :

$$P_{j,\gamma}(\mathrm{d}\mathbf{x}) = \frac{e^{\gamma' \mathbf{x}} \nu_j(\mathrm{d}\mathbf{x})}{\mathcal{L}_{\nu_j}(\gamma)}, \quad (4)$$

free of $\boldsymbol{\delta}$. Its moments are derivatives of the within-cell cumulant generating function: with $K_j(\gamma) = \log \mathcal{L}_{\nu_j}(\gamma)$, $\mathbb{E}_{P_{j,\gamma}}[\mathbf{x}] = \nabla K_j(\gamma)$ and $\text{Var}_{P_{j,\gamma}}(\mathbf{x}) = \nabla^2 K_j(\gamma)$.

The tilt (4) is precisely the change of measure that underlies saddlepoint approximation (Daniels, 1954; Jensen, 1995) and exponential-tilting estimators in the empirical-likelihood family (Efron, 1981; Kitamura and Stutzer, 1997; Schennach, 2007); here it appears not as an approximation device but as an exact algebraic identity for the score and information, as follows. Write, for each cell j ,

$$M_j^{(0)} = \sum_{i \in j} \omega_i, \quad \mathbf{m}_j = \sum_{i \in j} \omega_i \mathbf{x}_i, \quad \mathbf{Q} = \sum_{i=1}^N \omega_i \mathbf{x}_i \mathbf{x}_i', \quad (5)$$

and let $B_j = \sum_{i \in j} b'(\eta_i)$ and $R_j = \sum_{i \in j} (y_i - b'(\eta_i))$ denote the cell sums of means and of raw residuals.

Theorem 1 (Exact tilted-moment representation). *For the model (1)–(2) with canonical link, at every $\boldsymbol{\theta} \in \mathbb{R}^q$:*

(i) *The score is*

$$\phi \mathbf{U}(\boldsymbol{\theta}) = \begin{pmatrix} \sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'} \\ \sum_j \mathbf{w}_j R_j \end{pmatrix}, \quad \mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i) \mathbf{x}_i = B_j \mathbb{E}_{P_{j,\boldsymbol{\theta}}^{b'}}[\mathbf{x}],$$

where $P_{j,\boldsymbol{\theta}}^{b'}$ is the tilt of Definition 1 with b' in place of b'' .

(ii) *The information matrix is*

$$\phi \mathbf{F}(\boldsymbol{\theta}) = \begin{pmatrix} \mathbf{Q} & \sum_j \mathbf{m}_j \mathbf{w}_j' \\ \sum_j \mathbf{w}_j \mathbf{m}_j' & \sum_j M_j^{(0)} \mathbf{w}_j \mathbf{w}_j' \end{pmatrix}, \quad \mathbf{m}_j = M_j^{(0)} \mathbb{E}_{P_{j,\boldsymbol{\theta}}}[\mathbf{x}], \quad \mathbf{Q} = \sum_j M_j^{(0)} \mathbb{E}_{P_{j,\boldsymbol{\theta}}}[\mathbf{x} \mathbf{x}'].$$

(iii) *For the Poisson family, $b' = b''$, the two tilts coincide with the $\boldsymbol{\delta}$ -free Esscher transform (4), and $B_j = M_j^{(0)} = e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$, $\mathbf{m}_j = M_j^{(0)} \nabla K_j(\gamma)$, and the γ -block of $\mathbf{F}$ is $\sum_j M_j^{(0)} \{\nabla^2 K_j(\gamma) + \nabla K_j \nabla K_j'\}$.*

(iv) *The Newton step $\boldsymbol{\theta}^+ = \boldsymbol{\theta} + \mathbf{F}(\boldsymbol{\theta})^{-1} \mathbf{U}(\boldsymbol{\theta})$ computed from the quantities in (i)–(ii) is identical to the Newton step computed from the full $N \times q$ design.*

Claim (iv) is trivial once (i)–(ii) are established—the two computations produce the same matrices—but it is the operative point: the decomposition involves *no linearization, discretization, or approximation of any kind*, so the accelerated algorithm is not a fast approximate estimator but a fast exact organization of the arithmetic of the MLE. Section 5.1 exhibits agreement of the full iterate *sequences* at the 10^{-13} level, which is the accumulation of floating-point reordering only.

Statistically, the representation has content beyond bookkeeping. The δ -score in (i) compares each cell’s observed event mass with its expected mass under the current tilt; the cross-information in (ii) is the tilted covariance between cell-level and individual-level design; and the γ -block in (iii) decomposes into a within-cell tilted covariance $\nabla^2 K_j$ plus a between-cell term. In particular the identification of γ rests exactly on $\sum_j M_j^{(0)} \nabla^2 K_j(\gamma) \succ \mathbf{0}$: continuous covariates are identified through their *within-cell, tilt-weighted* variation—a fact that the block structure displays, and that drives the Schur-complement profile Hessians used throughout the companion theory in the online supplement. (When a continuous covariate is constant within cells—a route-level distance, say—it remains identified through between-cell variation as long as the cell-level design $\mathbf{w}_j$ is not saturated; Section 6.1 exercises this case.)

Corollary 1 (Per-iteration complexity). *One Newton iteration computed via Theorem 1 costs*

$$\underbrace{O\left(N(p^2 + p + 1)\right)}_{\text{microdata pass}} + \underbrace{O\left(J(k^2 + kp)\right)}_{\text{cell assembly}} + \underbrace{O\left((p + k)^3\right)}_{\text{solve}}$$

floating-point operations and one $O(N)$ memory sweep, versus $O(N(p + k)^2) + O((p + k)^3)$ for the full-design iteration. When $N \gg J(p + k)$ the leading-order saving is the factor $(p + k)^2/(p^2 + p + 1)$, which grows quadratically in the richness k of the categorical structure; when additionally $p = 0$ the microdata pass degenerates to the classical cell collapse and the entire iteration runs at $O(Jk^2)$.

Two remarks on the constant factors. First, the microdata pass is a fused stream: one evaluation of $b'(\eta_i)$, $b''(\eta_i)$ per observation, $p + 1$ multiply–accumulate reductions into cell bins, and a rank-one update of the $p \times p$ Gram matrix; on modern hardware it is memory-bandwidth-bound, so measured speedups (Section 5.1) are large but smaller than the FLOP ratio. Second, the accumulations are plain segmented sums—**bincount** operations—so they parallelize and stream trivially; no $N \times q$ design matrix is ever formed, which brings a problem whose dense design exceeds working memory (Section 6.2) within a laptop’s footprint.

3.2 Non-canonical links

So far the link has been canonical, $\theta_i = \eta_i$, where the Newton weight $b''(\eta_i)$ and the tilt coincide. The construction does not depend on this. For a general link g with $\mu_i = g^{-1}(\eta_i)$

the maximum-likelihood score for a linear exponential family is $\sum_i (y_i - \mu_i) \frac{\partial \mu_i / \partial \eta_i}{V(\mu_i)} \mathbf{a}_i$ and the expected (Fisher) information is $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$, with the standard IRLS weight and working residual

$$\omega_i = \frac{1}{V(\mu_i) g'(\mu_i)^2}, \quad r_i = (y_i - \mu_i) g'(\mu_i) \omega_i \quad (6)$$

(McCullagh and Nelder, 1989, Sec. 2.5), where V is the variance function and $g'(\mu) = \partial \eta / \partial \mu$. Comparing with Theorem 1, the only change is that the pair $(b'(\eta_i), b''(\eta_i)) = (\mu_i, \omega_i^{\text{can}})$ of the canonical case is replaced by (μ_i, ω_i) and the raw residual $y_i - \mu_i$ by r_i : the tilt is now weighted by the Fisher weight (6), and everything downstream—the cell sums, the Gram matrix, the block assembly—is unchanged.

Proposition 2 (Tilted Fisher scoring). *With ω_i and r_i as in (6), the expected-information scoring step admits the representation of Theorem 1, with $M_j^{(0)}, \mathbf{m}_j, \mathbf{Q}$ built from ω_i and $R_j = \sum_{i \in j} r_i$; the fixed points of the iteration are exactly the roots of the ML score equations, and Corollary 1 applies verbatim.*

Two points of intuition and scope. First, whether the moments must be *re-accumulated* each iteration depends on how ω_i depends on η_i . For canonical links $\omega_i = b''(\eta_i)$ moves with the iterate, so a fresh pass is needed—which is the pass the algorithm takes anyway. The gamma family with log link is the striking exception: $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i \equiv 1$, so the Fisher weight is constant, the “tilted” moments are ordinary exposure-weighted cell moments *fixed across iterations*, and after one pass each subsequent iteration costs $O(Np) + O(Jk^2)$ —the acceleration is at its most extreme precisely where the link is non-canonical. Second, the representation needs only that $\omega_i > 0$ and V, g be smooth on the range of the fitted means, so it covers the standard non-canonical GLMs—gamma or inverse-Gaussian with log link, binomial with probit or complementary log-log, Poisson with identity or square-root link—as expected-information scoring. It does not by itself guarantee that scoring *converges globally* for these links; that requires the log-concavity condition made precise for the absorbed-factor problem in Supplement Theorem S1, which holds for gamma–log and probit but not for every link. Observed information (true Newton) under a non-canonical link adds a term in b'''/g'' that is equally cell-decomposable but that we do not need, since Fisher scoring already delivers the MLE at the same per-iteration cost.

3.3 Families in scope

Three instances cover most applied use and are carried through the experiments. *Poisson with log link* (canonical): the tilt is the δ -free Esscher transform (4), and the event-history offset $o_i = \log t_i$ makes this the estimator of piecewise-constant hazard models on episode-split survival data (Holford, 1980; Whitehead, 1980). *Logistic* (canonical): the tilt weights $\omega_i = \mu_i(1 - \mu_i)$ concentrate the within-cell measure on observations near the decision boundary

Algorithm 1 Tilted-moment Newton / Fisher scoring

- 1: Precompute cell ids $j(i)$, cell design $\mathbf{W} \in \mathbb{R}^{J \times k}$, continuous columns $\mathbf{X} \in \mathbb{R}^{N \times p}$, offsets o_i ; initialize $\boldsymbol{\theta}_0 = (\boldsymbol{\gamma}_0, \boldsymbol{\delta}_0)$.
 - 2: **repeat**
 - 3: *Stream over* $i = 1, \dots, N$: compute $\eta_i = \boldsymbol{\gamma}'\mathbf{x}_i + (\mathbf{W}\boldsymbol{\delta})_{j(i)} + o_i$, the weight ω_i , and the residual r_i ; accumulate $M_{j(i)}^{(0)} += \omega_i$; $\mathbf{m}_{j(i)} += \omega_i\mathbf{x}_i$; $R_{j(i)} += r_i$; $\mathbf{U}_\gamma += r_i\mathbf{x}_i$; $\mathbf{Q} += \omega_i\mathbf{x}_i\mathbf{x}_i'$.
 - 4: *Cell assembly*: $\mathbf{U}_\delta = \mathbf{W}'\mathbf{R}$; $\mathbf{F}_{\gamma\delta} = \mathbf{m}'\mathbf{W}$; $\mathbf{F}_{\delta\delta} = \mathbf{W}'\text{diag}(M^{(0)})\mathbf{W}$; $\mathbf{F}_{\gamma\gamma} = \mathbf{Q}$.
 - 5: Solve the $(p+k)$ -dimensional system $\mathbf{F}\Delta = \mathbf{U}$; update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \lambda\Delta$ with backtracking $\lambda \in \{1, \frac{1}{2}, \dots\}$ on the log likelihood.
 - 6: **until** $\max_\ell |\Delta\theta_\ell|/(1 + |\theta_\ell|) < \tau$
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and depend on $\boldsymbol{\delta}$, so the moments are re-accumulated each iteration—which the single pass does anyway. *Gamma with log link* (non-canonical, Proposition 2): the Fisher weights are identically one, the moments are constant across iterations, and after the first pass each iteration costs $O(Np) + O(Jk^2)$. We deliberately stop the catalogue here; the point is the structure, not an enumeration.

4 Algorithms and implementation

4.1 The streaming pass

Algorithm 1 states the method operationally. Its preprocessing is one factorization of the categorical covariates into cell identifiers $j(i)$ and one construction of the $J \times k$ cell-level design $\mathbf{W}$; both are $O(N)$ given a sort or hash of the categorical columns, and both are shared across models on the same cells.

By Theorem 1(iv) the iterate sequence is identical to full-design Newton from the same start; every design decision below concerns constants, not correctness.

4.2 Numerical safeguards

Three matter in practice. (i) *Monotonicity*: the backtracking line search on the log likelihood guarantees ascent and terminates, since the Newton direction of a positive-definite system is an ascent direction; in well-conditioned problems the full step is accepted after the first iterations and convergence is quadratic. (ii) *Log-domain evaluation*: likelihood and mean evaluations use `log1p` / `logaddexp`-style kernels so that extreme linear predictors (produced, e.g., en route to a separated fit) yield $-\infty$ log likelihoods rather than overflow. (iii) *Rank deficiency*: if the assembled $\mathbf{F}$ is singular—collinear cell-level columns, or vanishing weights under quasi-separation—the solve falls back to a minimum-norm (pseudoinverse) direction and the iteration runs to its cap with the non-convergence flagged, never silently; separation is a

property of the data, not of the algorithm, and Section S3 of the supplement develops the Firth-adjusted variant that restores a finite maximizer inside the same pass.

4.3 Memory layout and parallelism

The gap between measured and FLOP ratios merits a constructive comment, since closing it is an implementation exercise with a known playbook. The tilted pass streams $O(Np)$ numbers and performs $O(Np^2)$ flops, so for small p its arithmetic intensity is low and the memory bus, not the ALU, sets the speed; the dense baseline, by contrast, is a BLAS-3 Gram update running near peak. Three standard remedies raise the tilted pass toward bandwidth saturation and beyond: (i) *cache locality*: storing the microdata sorted by cell converts the binned accumulations into contiguous segmented sums, so each cell’s partial moments stay in registers/L1 and the pass touches each cache line once (this is how the Mata implementation in `stpois` organizes the sweep, following [Correia, 2016](#)); (ii) *fusion*: the weight evaluation, residual, binned sums, and Gram update share one loop, so the data are read once per iteration rather than once per moment; and (iii) *parallelism*: cells are independent, so the pass shards trivially across threads with per-thread partial moments and a Jk -sized reduction, with no atomics on the hot path—the same structure GPU segmented-reduction primitives expect. The single-threaded NumPy reference implementation uses none of (i)–(iii) beyond what `bincount` provides, so the reported speedups should be read as a floor; the FLOP ratio governs how the comparison scales with (p, k) , and the bandwidth analysis governs the attainable constant.

The *memory* footprint is where the reorganization is most decisive, and it is easy to state exactly. The tilted pass holds the loaded columns, $O(N(p + 1))$ doubles, plus the accumulators: J cell scalars, a $J \times p$ moment array, and the $(p + k) \times (p + k)$ information— $O(N(p + 1)) + O(J(k + p) + (p + k)^2)$ in total, with no term in Nk . A dense design is $N(p + k)$ doubles, and a general-purpose IRLS adds working copies on top; the measured peak resident sets bear this out (Tables 2 and 5): ~ 80 MB for the tilted fit against ~ 1 GB dense and ~ 4 GB for `statsmodels` at $N = 2 \times 10^5$, widening to 3.8 GB versus a 200 GB dense design at $N = 5 \times 10^7$. Because peak memory, not FLOPs, is what makes a register-scale fit feasible or not, this is the practically load-bearing comparison. Preprocessing—the one-time sort or hash that builds the cell identifiers and, for absorbed factors, the group indices—is $O(N \log N)$ or $O(N)$ and is shared across all models fitted on the same cells (a sweep of specifications, a bootstrap, a cross-validation), so it is reported separately from the per-iteration cost throughout and is amortized whenever more than one model is fitted on a design.

Reproducibility of the timings. All wall-clock and peak-memory figures are measured on a single core of a commodity x86-64 workstation (AVX2, 32 KiB L1 / 1 MiB L2 per core, shared L3), with each competitor fitted in a fresh process so that peak resident set isolates its footprint; times are medians over repeated fits. The reference implementation is pure NumPy/SciPy

on the system BLAS, single-threaded (BLAS threads pinned to one) so that the comparison reflects algorithmic organization rather than parallel speedup, which Section 4.3 argues the tilted pass admits but does not yet use. The exact hardware affects the constants, not the ratios or the $O(\cdot)$ trends, which are the paper’s claims; every figure regenerates from the scripts in the code supplement.

4.4 Robust and clustered variance estimation

All variance estimators used in practice share the cell structure. The observed/expected information is $\mathbf{F}$ itself (Theorem 1). The HC-type sandwich needs $\sum_i \hat{u}_i^2 \mathbf{a}_i \mathbf{a}_i'$ with $\hat{u}_i = y_i - \hat{\mu}_i$; its blocks accumulate cellwise exactly like (5) with ω_i replaced by $\hat{u}_i^2$. The cluster-robust estimator sums scores within clusters before the outer product; when clusters are unions of cells (the common design: cluster on region, cells within region) the accumulation is again one pass. These identities are what make *inference*, not just point estimation, run at cell-collapsed cost; they are implemented in the reference code and validated against dense computation in the test suite.

4.5 Absorbed fixed-effect factors

Applications often add factors whose level counts preclude even cell-level treatment—firms, hospitals, municipality–year interactions. The framework composes with the standard absorption device: augment the linear predictor with G absorbed factors $\alpha_{g(i)}^{(g)}$, concentrate them out by cycling closed-form profile updates (for Poisson, $\alpha_a^{(g)} = \log\{\sum_{i:g(i)=a} y_i / \sum_{i:g(i)=a} \exp(\eta_i - \alpha_a^{(g)})\}$), the multiplicative update of Bergé, 2018; Correia et al., 2020), and take each tilted-moment Newton step on $(\boldsymbol{\gamma}, \boldsymbol{\delta})$ at the profiled effects, with the inner cycle re-solved at every trial point of the line search (Algorithm S1 of the supplement). Each inner sweep is $O(N)$; with a single absorbed factor the inner solve is exact in one sweep.

This inner–outer scheme is exactly the algorithm used across the HDFE–GLM software ecosystem, and it is the one for which the applied literature has recorded no convergence guarantee. Because the algorithm is central to the absorbed-factor results of this paper, we state the guarantee here and defer its proof, and the finer rate and asymptotic theory, to the online supplement.

Theorem 2 (Global convergence of the absorbed-factor iteration). *Suppose the link is canonical, or more generally log-concave in the sense that each per-observation contribution $\eta \mapsto \ell_i(\eta)$ is strictly concave; suppose the maximum-likelihood estimate exists (no separation, which the drop rule below enforces for the absorbed blocks); and suppose the non-absorbed information block is nonsingular on the initial superlevel set. Then the iterates of the inner–outer scheme above—closed-form or exact one-dimensional block maximizations of the absorbed effects, alternated with Armijo-damped tilted-moment Newton/scoring steps for $(\boldsymbol{\gamma}, \boldsymbol{\delta})$ —increase the likelihood*

monotonically and converge to the maximum-likelihood estimate, from any starting value, for any (essentially) cyclic ordering of the blocks.

The proof (Supplement Theorem S1) is a block-coordinate-ascent argument on the concave profile objective, handling the parameter space’s flat directions and the inexact (damped-Newton) outer block; the log-concave-link generality covers gamma–log and probit as well as the canonical families. The supplement further gives the *local rate* (Theorems S2–S3): for two absorbed factors the per-sweep contraction equals $\cos^2$ of the Friedrichs angle between the weighted factor subspaces—the squared second singular value of a normalized weighted cross-tabulation—and for $G \geq 3$ it is the spectral radius of a level-space sweep operator built from the pairwise cross-tabulations alone, computable by power iteration before touching the microdata, with an angle-based product certificate; observed contraction factors match these predictions to three decimals. Because the estimator *is* the MLE, its asymptotics are the MLE’s: classical at fixed dimension, and, in the proportional regime of one Poisson fixed effect per bounded-size cell ($J/N \rightarrow c \in (0, 1)$), unbiased with valid cell-clustered standard errors, because profiling then coincides with conditioning (Theorems S4–S6). Separated absorbed groups (no events) are detected and dropped a priori, as in [Correia et al. \(2020\)](#); Section 6.2 exercises the full composition at scale, and Section 6.1 checks it against `pyfixest` on real data.

5 Numerical experiments

All experiments run on a single laptop-class core (Python/NumPy reference implementation; scripts `sim1*` in the code supplement) and complete in minutes; every number below regenerates from a fixed seed. The studies validating the supplement’s absorption, asymptotic, and separation theory (convergence rates for two and three factors, the proportional regime, Firth under quasi-separation, the logit bias and its conditional repair) are reported in Section S4 of the online supplement.

5.1 Exactness and per-iteration cost

Study 1 fits Poisson, logistic, and gamma models with $J \in \{50, 200, 500\}$ cells, $p \in \{2, 5\}$ continuous covariates, and N up to 10^6 , tracing the full iterate sequence of tilted-moment Newton and of textbook dense Newton from the same starting point. Table 1 reports the maximum deviation between the two iterate sequences across all iterations, the final-estimate deviation, and the per-iteration wall-clock times.

Three observations. First, the iterate sequences agree to 10^{-13} – 10^{-15} throughout—floating-point reordering noise—confirming Theorem 1(iv): the tilted iteration is the same algorithm as full-design Newton, not an approximation to it. Second, measured per-iteration speedups range from $12\times$ ($J = 50$) to $117\times$ ($J = 500$), growing with k as Corollary 1 predicts, though below

Table 1: Study 1: tilted-moment versus dense Newton. “Iterate dev.” is the maximum absolute difference between the two iterate sequences over all iterations and coordinates; times are medians per Newton iteration. Selected configurations; the full grid is in the supplement.

Family	N	J	p	Iterate dev.	Final dev.	t_{dense} (s)	t_{tilted} (s)
Poisson	2×10^5	50	2	5.1×10^{-14}	2.4×10^{-15}	0.119	0.010
Poisson	2×10^5	200	2	5.1×10^{-14}	3.7×10^{-15}	0.503	0.010
Poisson	2×10^5	500	2	6.0×10^{-13}	3.0×10^{-14}	1.481	0.017
Poisson	10^6	200	2	7.4×10^{-14}	1.1×10^{-14}	2.490	0.062
Poisson	2×10^5	200	5	1.1×10^{-13}	1.9×10^{-15}	0.520	0.016
Logistic	2×10^5	500	2	2.2×10^{-14}	4.0×10^{-15}	1.738	0.015
Logistic	10^6	200	2	8.2×10^{-15}	2.2×10^{-16}	2.918	0.082
Gamma	2×10^5	500	2	1.3×10^{-13}	5.2×10^{-15}	1.443	0.013
Gamma	10^6	200	2	2.5×10^{-13}	1.3×10^{-15}	2.662	0.058

the raw FLOP ratios ($430\times$ – $4700\times$) because the dense baseline runs BLAS-3 while the tilted pass is memory-bound—the FLOP ratio is the right guide to scaling, the measured ratio to wall-clock budgeting. Third, the dense baseline at $N = 10^6$, $J = 200$ already needs a 1.6 GB design matrix where the tilted pass needs the p continuous columns only; at the dimensions of Section 6.2 the dense fit exceeds the memory of a typical workstation before the first iteration.

5.2 Comparison with production implementations

Study 1b benchmarks total time and peak memory to convergence (identical tolerance 10^{-9} , comparable starting values) against the implementations a practitioner would actually reach for. Each fit runs in a fresh process so that peak resident set size (RSS) isolates one method’s footprint; times are medians over repeated fits and iteration counts are reported alongside. For the non-absorbed model (Table 2, $N = 200,000$, $J = 200$, $p = 2$) the competitors are our own dense Newton; **statsmodels** (Seabold and Perktold, 2010), the standard general-purpose Python GLM (dense IRLS); a sparse-design IRLS holding the design (continuous columns plus cell dummies) in compressed-sparse-row form and forming and solving the normal equations sparsely—the “the dummies are sparse, so use a sparse solver” route a careful analyst would code; and **glum** (Balzer et al., 2021), QuantCo’s high-performance GLM, which carries categorical covariates in a dedicated split-matrix backend and is among the fastest Python GLM implementations.

Every method returns the same coefficients to 10^{-15} : the comparison is purely one of arithmetic organization and memory. The tilted pass is fastest and, more decisively, lightest: its ~ 80 MB is the loaded columns plus cell accumulators, against ~ 1 GB for a dense design and ~ 4 GB for **statsmodels**’ internal copies, a 12–50 $\times$ memory advantage that grows with N (Section 5.4). The sparse-design IRLS is the strongest competitor at $p = 2$, as it should be—cell dummies are the textbook sparse design—but the second panel of the study, which sweeps

Table 2: Study 1b, no absorbed factor: total time to convergence, peak resident memory, and iterations, with the maximum absolute coefficient deviation from the tilted fit. $N = 200,000$, $J = 200$ cells, $p = 2$, tolerance 10^{-9} ; every method computes the same MLE. (t_{tilted} shown is the warm value; the very first subprocess spawned in a run pays a one-time interpreter/BLAS start-up cost.)

Family	Method	Time (s)	Peak (MB)	Iters	Max. dev.
Poisson	tilted	0.08	80	9	—
	sparse IRLS	0.26	107	9	6×10^{-17}
	glum	0.90	227	7	3×10^{-17}
	dense Newton	3.1	998	8	6×10^{-17}
	statsmodels	14.2	3972	7	2×10^{-15}
Logistic	tilted	0.10	76	8	—
	sparse IRLS	0.23	107	8	3×10^{-17}
	glum	0.37	229	5	2×10^{-16}
	dense Newton	3.1	997	8	6×10^{-17}
	statsmodels	14.0	3972	7	6×10^{-15}

Table 3: Study 1b, continuous-dimension sweep (Poisson, $N = 200,000$, $J = 200$): time and peak memory for the exact-MLE methods. The tilted cost grows in p on the continuous block only; the sparse Gram fill-in makes the sparse route super-linear in p .

p	Time (s)			Peak (MB)		
	tilted	dense	sparse	tilted	dense	sparse
2	0.08	3.0	0.27	80	998	107
20	0.41	3.4	2.73	135	1081	262
50	1.18	4.4	12.5	227	1219	512

the continuous dimension $p \in \{2, 20, 50\}$ (Poisson; Table 3), shows where the two part ways: the tilted cost grows as $O(Np^2)$ on the dense continuous block only, while the sparse route’s Gram matrix fills in and its factor grows super-linearly, so at $p = 50$ the sparse solver is $10\times$ slower than tilted and dense Newton overtakes it. **statsmodels** and **glum** are omitted from the p -sweep: the former needs tens of seconds and gigabytes per fit, and **glum**’s unpenalized ($\alpha = 0$) solver returned a singular-matrix error for $p \geq 20$ —it is engineered for penalized fitting, and the exact MLE is outside its robust regime.

5.3 High-dimensional fixed effects: comparison with **pyfixest**

When a high-dimensional factor is absorbed rather than entered as dummies, the relevant competitor is a dedicated HDFE–GLM tool. Study 1d (Table 4) benchmarks the absorbed-factor path against **pyfixest** (Bergé, 2018), the Python implementation of the **fixest** alternating-projections algorithm, on Poisson models with $N \in \{2, 5\} \times 10^5$, $J = 100$ cells, $p = 2$, and one absorbed factor of 500 to 20,000 levels; dense-dummy Newton and **glum** with dense dummies

Table 4: Study 1d, one absorbed factor: time, peak memory, and deviation from the tilted fit. Dense-dummy Newton and `glum` form the $N \times (J + \text{levels})$ design explicitly and fail (singular Hessian, or out of memory / did not finish within 20 minutes) once the factor exceeds a few thousand levels. `pyfixest` matches to its own demeaning tolerance ($\sim 10^{-4}$).

Levels	Method	Time (s)	Peak (MB)	Max. dev.
500	tilted	0.10	79	—
	<code>pyfixest</code>	0.19	378	4×10^{-4}
	dense Newton	10.0	2837	4×10^{-13}
	<code>glum</code>	singular		—
5,000	tilted	0.16	79	—
	<code>pyfixest</code>	0.21	372	5×10^{-4}
	dense Newton	out of memory		—
	<code>glum</code>	singular		—
20,000	tilted	0.36	112	—
	<code>pyfixest</code>	0.64	544	4×10^{-4}
	dense Newton	out of memory		—
	<code>glum</code>	out of memory		—

are included as references.

Against the specialized tool the tilted absorption is roughly twice as fast and four to five times lighter, while returning the exact MLE (`pyfixest` agrees to its demeaning tolerance). The dense-dummy routes confirm why HDFE tools exist at all: they exhaust memory once the factor passes a few thousand levels. The like-for-like comparison on *real* data in Section 6.1, where the two agree to 10^{-9} , is the more stringent test; the point of the synthetic sweep is the memory and timing trend.

5.4 Scaling in N and memory footprint

Study 1e runs the tilted Poisson fit at N from 10^6 to 5×10^7 ($J = 500$, $p = 2$), each in a fresh process (Table 5). Wall-clock is linear in N at ≈ 1.2 seconds per million records once the fixed start-up cost is amortized, and peak memory is linear as well—the loaded columns $O(N(p+1))$ plus the $O(J(k^2 + kp))$ accumulators—reaching 3.8 GB at $N = 5 \times 10^7$. A dense $N \times (p+J)$ design for the same problem would occupy 200 GB, a factor of 53 more; the tilted pass forms no design matrix, so the whole fit stays within a laptop’s memory at a scale where the dense and sparse routes cannot allocate their first iterate. This is the controlled counterpart of the register-scale application (Section 6.2): the same $O(N)$ time and memory, measured across a two-order-of-magnitude range of N .

Table 5: Study 1e: tilted Poisson fit scaling ($J = 500$, $p = 2$). Time is linear in N ; peak memory is linear and far below the size of a dense design, which for $N = 5 \times 10^7$ would need 200 GB.

N	Time (s)	Peak (GB)	Dense design (GB)	Ratio
1×10^6	1.4	0.17	4.0	24
5×10^6	6.9	0.44	20.1	46
2×10^7	23.6	1.56	80.3	51
5×10^7	59.6	3.79	200.8	53

5.5 Sensitivity: imbalance and rare events

Study 1f re-runs the exactness comparison under stress: cell sizes drawn from a Zipf(1.6) law (a few giant cells, a long tail of tiny ones) and event rates pushed to $\sim 0.2\%$ (Poisson means ~ 0.002 ; logistic prevalence $\sim 0.5\%$), with event-free cells dropped as the MLE requires. Across all eight family $\times$ scenario combinations the maximum deviation from dense Newton stays at 10^{-15} – 10^{-14} , convergence takes at most 12 iterations (rare-event fits need a few more steps, as their likelihoods are flatter), and the streaming pass time is unchanged: the accumulation is a data-independent $O(N)$ sweep, so imbalance redistributes work across cells without changing its total. Exactness is an algebraic identity, so it is expected to survive stress; the study confirms that the *implementation*—binned accumulation, line search, drop rules—inherits it.

5.6 Numerical stability

Because the tilted iteration is the same Newton/Fisher iteration as the full-design fit (Theorem 1), the question a numerical analyst asks is whether the *implementation*—binned accumulation, log-domain likelihood, line search, pseudoinverse fallback—inherits the conditioning of the underlying problem or degrades it. Study 1g (Table 6) answers this on a battery of adversarial scenarios at $N = 10^5$: near-collinear covariates (correlation 0.9999); an ill-scaled covariate (multiplied by 10^6 , giving information-matrix condition number $\sim 10^{17}$); extreme offsets spanning 30 log units; linear predictors driven to ± 40 (over/underflow territory); Zipf(1.3)-unbalanced cells; and an informative covariate carried on a 10^8 pedestal (catastrophic cancellation in any naive sum of squares). For each we report the condition number, the deviation from dense Newton on the same data, and the deviation from a cancellation-free reference—the mathematically identical model with the continuous columns standardized, then mapped back—so that a genuine conditioning failure is separated from a faithful reproduction of a hard problem.

The reading is clean. At condition numbers up to 10^{17} , with linear predictors at ± 40 , exposures spanning 30 log units, near-collinear covariates, and Zipf-unbalanced cells, the tilted fit reproduces dense Newton and the standardized reference to 10^{-11} – 10^{-17} : the accumulation neither introduces nor amplifies ill-conditioning. The one genuine failure—an informative signal

Table 6: Study 1g: stability battery (Poisson; the logistic panel is similar). “ κ ” is the information-matrix condition number; “vs. dense” the max coefficient deviation from dense Newton on the raw data; “vs. std” the deviation from the standardized-covariate reference (a conditioning diagnostic). The tilted fit matches dense and the well-conditioned reference throughout, except under a 10^8 pedestal, where dense diverges *identically* (catastrophic cancellation shared by every double-precision solver) and centering the covariate—the one-line standardization the paper recommends—restores full accuracy.

Scenario	κ	vs. dense	vs. std	converged
balanced	4×10^5	2×10^{-14}	6×10^{-17}	yes
collinear (0.9999)	5×10^5	8×10^{-14}	1×10^{-13}	yes
ill-scaled ($\times 10^6$)	3×10^{17}	4×10^{-14}	0	yes
extreme offset	3×10^5	1×10^{-15}	6×10^{-17}	yes
huge η (± 40)	2×10^{11}	1×10^{-11}	3×10^{-17}	yes
unbalanced (Zipf)	3×10^5	1×10^{-15}	6×10^{-17}	yes
10^8 pedestal (raw)	9×10^{29}	1×10^7	0.27	no
10^8 pedestal (centered)	3×10^5	5×10^{-14}	3×10^{-17}	yes

on a 10^8 pedestal—is a catastrophic-cancellation hazard of the raw parametrization, not of the tilted pass: dense Newton diverges by the same 10^7 , and standardizing the covariate (which the reference implementation offers and the paper recommends as routine preprocessing, following standard practice; [Higham, 2002](#)) returns the fit to full accuracy. We flag this explicitly rather than suppress it: no double-precision GLM solver survives a covariate whose signal sits 16 digits below its magnitude, and the honest remedy is to standardize, not to claim immunity.

6 Applications

6.1 On-time performance of NYC flights

The first application is to public data, chosen so that every number can be reproduced from a fresh download: the `nycflights13` data set ([Wickham, 2021](#))—all 336,776 flights departing New York City’s three airports in 2013, from the U.S. Bureau of Transportation Statistics. We model the probability of a late arrival (arrival delay ≥ 15 minutes, the FAA on-time criterion; 24.5% of the 327,346 flights with an observed delay) by logistic regression with a cell structure of carrier $\times$ origin $\times$ destination $\times$ month— $J = 3,863$ observed combinations, entering through a main-effects cell-level design with $k = 131$ parameters (16 carriers, 3 origins, 103 destinations, 12 months)—and $p = 2$ continuous covariates: scheduled departure time (in hours, centered) and log distance (standardized). One destination (LEX, a single on-time flight) is quasi-separated—its dummy’s MLE is $-\infty$ —and is dropped by the level-wise separation rule, after which the MLE exists.

The tilted logistic fit converges in 7 Newton iterations, 1.3 seconds including robust standard errors. The diurnal effect is the substantive headline: the odds of a late arrival multiply by

1.11 per hour of scheduled departure time (robust SE on the log scale 0.001)—delays cascade through the day, a finding familiar from the air-transportation literature, here estimated jointly with a full operational cell structure. Log distance illustrates the boundary case flagged in Section 3: distance is constant within origin–destination routes, hence within cells, so after destination main effects it is identified only through cross-route contrasts and is correspondingly imprecise (+0.65, robust SE 0.28)—the estimator handles cell-constant continuous covariates correctly, and the standard error reports how little such a covariate adds.

As external validation, the identical model fitted with `statsmodels` GLM on the dense $327,345 \times 133$ design (0.35 GB) gives a maximum absolute coefficient deviation of 2.3×10^{-13} , at 19 seconds versus 1.3: the same estimator, $15\times$ cheaper, with no design matrix in memory.

A second model on the same data exercises the high-dimensional absorption path against `pyfixest`. We now absorb *aircraft*—the 3,692 tail numbers that survive the separation drop—crossed with a twelve-level month cell structure (an aircraft flies across months, so the two factors are identified), keeping the two continuous covariates. The tilted logistic fit with absorbed aircraft converges in 31 iterations and 1.5 seconds; `pyfixest`’s `feglm` on the same specification takes 2.1 seconds, and the two agree on the continuous coefficients to 1.2×10^{-9} (departure hour 0.103994 from both). This is the paper’s most stringent external check: on real high-dimensional data, the tilted absorption reproduces the estimate of a mature, purpose-built HDFE–GLM package to nine digits, at comparable speed, while also carrying the cell-structured part of the model that `pyfixest` treats as ordinary covariates (script `app_flights.py`; the data file is fetched from the public repository cited in the script).

6.2 A register-scale stress test

The second application is a stress test at the scale that motivates the framework: spatial epidemiology on an emulated population register. Because genuine Nordic register microdata cannot leave certified secure environments, the study uses a fully reproducible synthetic register, calibrated to published all-cause mortality gradients (Gompertz age slope, female advantage of ~ 0.45 on the log scale, educational gradients of 0.1–0.4, a secular decline of 1.2% per year); the generator is 40 lines of the supplement and every number below regenerates end to end in under a minute. The scientific template is a standard one—does a within-municipality environmental exposure predict mortality after fine demographic and spatial-temporal control?—and the computational shape combines every element of Sections 2–4.

The data comprise $N = 5,000,000$ person-year episodes (35,526 deaths; crude rate 9.5 per 1,000 person-years) with: municipality $\times$ year fixed effects ($500 \times 20 = 10,000$ groups) absorbed by the profile Gauss–Seidel scheme of Section 4.5 (Algorithm S1); a cell-level design of education (4) $\times$ sex (2) $\times$ age band (12), $J = 96$ cells, $k = 16$ parameters; and $p = 2$ continuous covariates—standardized log income and a standardized pollution-exposure index correlated with the municipality effects (making the fixed effects non-ignorable for it). Standard errors

are clustered on the 500 municipalities.

The fit: 322 zero-event municipality–year groups (159,464 episodes) are dropped by the separation rule of Section 4.5; the remaining model—9,678 absorbed effects, 16 cell-level and 2 individual-level coefficients—converges in 8 tilted Newton iterations, 11.2 seconds end to end on one core. A dense dummy-variable design for the same model would occupy $N \times 9,696$ doubles ≈ 0.4 TB before the first iteration. For verification at feasible scale, the subpopulation of the first 10 municipalities ($n = 97,723$, 196 absorbed groups) was fitted both ways: maximum absolute coefficient deviation 4.0×10^{-11} , tilted 0.1s versus dense 3.0s.

Substantively, the model recovers the calibrated truth within sampling error: income -0.180 (cluster SE 0.005; truth -0.18), pollution $+0.049$ (SE 0.005; truth $+0.06$, within two SEs, identified from within-municipality variation only, as it must be after absorption), female -0.455 (truth -0.45), education gradients $-0.13/ -0.27/ -0.40$ (truth $-0.12/ -0.25/ -0.40$), and a monotone age gradient tracking the Gompertz truth to two decimals. The point of the exercise is not these numbers—the generator made them true—but the demonstration that a model of this shape, with inference, is an interactive-latency computation under the tilted-moment organization, whereas a dense dummy-variable design for the same model would need to allocate a terabyte-scale matrix before its first iteration; on genuine register data inside a secure environment, where wall-clock and memory budgets are tight and software dependencies restricted, this difference is often what determines whether such models are fitted at all.

7 Discussion

The paper’s message is one computational idea, developed to the point of routine use: when a GLM mixes a rich categorical structure with a few continuous covariates, the exact score and information—hence exact Newton, exact profile Hessians, and exact robust and clustered covariance matrices—are functions of three tilted within-cell moments that one streaming pass produces. Ordinary collapse fails because the likelihood surface needs the whole within-cell distribution (Proposition 1); estimation succeeds because each iteration needs only the current tilt of that distribution (Theorem 1). The experiments quantify what this buys over the implementations in actual use, including the strongest fair competitor (a sparse-design IRLS), and the applications show the method doing routine work on public and register-scale data.

The device composes with, rather than competes against, the fixed-effect absorption stack: absorbed factors handle dummy structure too large even for cells, and the tilted pass handles everything else. The companion theory for that composition—global convergence of the nonlinear alternating-projections cycle for log-concave links, exact local rates for any number of absorbed factors from pairwise weighted cross-tabulations, asymptotics up to the proportional regime $J/N \rightarrow c$ with its Poisson unbiasedness and its logit failure-and-repair, and Firth and

penalized variants computed inside the same pass—is developed, proved, and validated in the online supplement; we kept it out of the main text so that the computational contribution stands on its own terms.

Software. The algorithms are implemented twice. The Stata command `stpois` implements the Poisson event-history case in production form (its software article documents the interface); a self-contained Python/NumPy reference implementation (`tilted_glm.py`, roughly 800 lines), which generated every number in this paper and the supplement, accompanies the manuscript and is organized to make the correspondence between code and results line-by-line auditable. Both are available at <https://github.com/andreasjungstrm/stpois>.

Acknowledgments. The author thanks colleagues at SOFI for discussions of register-based event-history workflows that motivated this work.

A Proofs for Section 2

A.1 Proof of Proposition 1

(i) If $\mathbf{x}_i = \bar{\mathbf{x}}_j$ within cells, then $\eta_i = \boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j + \log t_i$ and (3) becomes $\sum_j \{(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j)D_j - T_j e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j}\} + \text{const}$, a function of the displayed cell statistics; the collapsed likelihood is the microdata likelihood up to an additive constant, so their maximizers and Hessians coincide.

(ii) For the Poisson family,

$$\ell(\boldsymbol{\gamma}, \boldsymbol{\delta}) = \sum_j \left\{ \boldsymbol{\gamma}'S_j + \boldsymbol{\delta}'\mathbf{w}_j D_j - e^{\boldsymbol{\delta}'\mathbf{w}_j} \mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \right\} + \text{const},$$

which establishes computability from $\{S_j, D_j, \mathcal{L}_{\nu_j}\}$. Conversely, suppose a statistic permits evaluation of ℓ on an open set $\mathcal{O} \ni (\boldsymbol{\gamma}_0, \boldsymbol{\delta}_0)$. The functions $\boldsymbol{\delta} \mapsto e^{\boldsymbol{\delta}'\mathbf{w}_j}$ for distinct vectors $\mathbf{w}_j$ are linearly independent on any open set, so from $\boldsymbol{\delta} \mapsto \sum_j e^{\boldsymbol{\delta}'\mathbf{w}_j} \mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$ one recovers each $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$ for each $\boldsymbol{\gamma}$ in the open slice. Each ν_j is a finite discrete measure, so $\mathcal{L}_{\nu_j}$ is entire; its values on an open subset of $\mathbb{R}^p$ determine it everywhere by analytic continuation, and the Laplace transform of a finite (positive) measure determines the measure. Hence the statistic determines ν_j , whose minimal description has dimension equal to (one plus p times) its support size.

(iii) Jensen’s inequality applied to the strictly convex map $z \mapsto e^z$ under the probability measure ν_j/T_j gives $T_j^{-1} \mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \geq \exp(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j)$ with equality iff $\boldsymbol{\gamma}'\mathbf{x}$ is ν_j -a.s. constant. $\square$

A.2 Proof of Theorem 1

Differentiating (3) with $\theta_i = \eta_i$: $\phi \partial \ell / \partial \boldsymbol{\gamma} = \sum_i (y_i - b'(\eta_i)) \mathbf{x}_i$ and $\phi \partial \ell / \partial \boldsymbol{\delta} = \sum_i (y_i - b'(\eta_i)) \mathbf{w}_{j(i)}$. Grouping the second sum by cells and using that $\mathbf{w}_j$ is constant within cell j gives $\sum_j \mathbf{w}_j R_j$;

grouping the first gives $\sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'}$ with $\mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i) \mathbf{x}_i = B_j \mathbb{E}_{P_{j,\theta}^{b'}}[\mathbf{x}]$ by the definition of the b' -tilt. Differentiating again, $\phi \mathbf{F} = \sum_i b''(\eta_i) \mathbf{a}_i \mathbf{a}_i'$ with $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$; expanding the blocks and grouping by cells yields (ii), and the moment identities follow from Definition 1. For (iii), $b' = b'' = \exp$ gives $\omega_i = e^{\delta' \mathbf{w}_j} t_i e^{\gamma' \mathbf{x}_i}$, the cell factor cancels in the normalization, and $M_j^{(0)} = e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$; the cumulant identities $\mathbb{E}_{P_{j,\gamma}}[\mathbf{x}] = \nabla K_j$ and $\text{Var} = \nabla^2 K_j$ are the standard derivatives of a log-Laplace transform, and $\mathbb{E}[\mathbf{x} \mathbf{x}'] = \text{Var} + \mathbb{E} \mathbb{E}'$ gives the stated γ -block. (iv) The score and information assembled from (i)–(ii) are, entry for entry, the same matrices as those computed from the stacked design; hence so is the Newton step. $\square$

A.3 Proof of Corollary 1

The microdata pass evaluates, per observation, η_i from the cached cell linear predictor plus $\gamma' \mathbf{x}_i$ ($O(p)$), the scalars $b'(\eta_i), b''(\eta_i)$ ($O(1)$), the binned accumulations of ω_i and $\omega_i \mathbf{x}_i$ into cell $j(i)$ ($O(p)$), and the symmetric rank-one update $\omega_i \mathbf{x}_i \mathbf{x}_i'$ ($O(p^2)$ scalar multiply-adds, $p(p+1)/2$ distinct); total $O(N(p^2 + p + 1))$. Assembly of $\sum_j M_j^{(0)} \mathbf{w}_j \mathbf{w}_j'$ and $\sum_j \mathbf{m}_j \mathbf{w}_j'$ costs $O(J(k^2 + kp))$, and the symmetric solve $O((p+k)^3)$. The dense iteration forms $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ at $O(N(p+k)^2)$ plus the same solve. Ratios follow. With $p = 0$ the pass reduces to two binned sums ($R_j, M_j^{(0)}$), and if the offsets enter through cell sums (pure rate models), even these are fixed after one pass. $\square$

A.4 Proof of Proposition 2

The ML score for a general link is $\sum_i (y_i - \mu_i) \frac{\partial \mu_i / \partial \eta_i}{V(\mu_i)} \mathbf{a}_i = \sum_i (y_i - \mu_i) g'(\mu_i) \omega_i \mathbf{a}_i = \sum_i r_i \mathbf{a}_i$, using $\partial \mu / \partial \eta = 1/g'(\mu)$, and the expected information is $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ (McCullagh and Nelder, 1989, Sec. 2.5). These have the same cell-decomposable form as in Theorem 1 with (ω_i, r_i) in place of $(b''(\eta_i), y_i - b'(\eta_i))$, so the accumulation and complexity claims carry over. A fixed point of the scoring iteration has $\sum_i r_i \mathbf{a}_i = \mathbf{0}$, i.e., is a root of the ML score equations. For the gamma-log pair, $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i = 1/(\mu_i^2 \mu_i^{-2}) = 1$; the moments (5) are then constant in θ , and only $R_j = \sum_{i \in j} (y_i - \mu_i)/\mu_i$ requires re-evaluation, an $O(Np)$ pass. $\square$

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